Solution overview: Credit Risk Portfolio Monitor

Streamlit dashboard over 5 synthetic bank tables (14,685 loan accounts, 10,000 customers). Repo layout and run commands are in README.md.

What the dashboard shows and how an analyst uses it

Portfolio	KPIs; exposure and actual default rate by model risk band (A-E); exposure by loan type x band; default rate, PD and exposure by any dimension; cross-cut heatmap (any two dimensions); vintage and month-on-book trends; drill-down from a segment to its accounts. Sidebar filters (loan type, region, employment, income, band, year) apply throughout. Use: find where risk and money concentrate, then drill in.
Account	PD, band, percentile, top drivers (per-account SHAP), profile, 12-month EMI/DPD chart, bureau lines, collections, anomaly reasons. Use: explain a score before a limit, collections or override decision.
Ask the Portfolio	Plain-English questions with exact figures, the scope used, the result table and the query spec.
Anomaly review	Prioritised queue of flagged accounts, why each was flagged, peer comparison, decision + log.
Model / Data	Metrics, ROC, deciles, importance, monitoring plan; join logic and full data-quality log.

The two AI features

	Ask the Portfolio (LLM)	Anomaly detection
How it works	LLM call 1 turns the question into a JSON query using only the schema (no rows). Code validates it against a whitelist and computes the answer with pandas. LLM call 2 words a 2-3 sentence answer from the aggregated result; every number is checked against the result, otherwise a template answer is used. Out-of-scope questions (forecasts, 'why', outside facts, personal data) are declined. Works without a key via a keyword parser.	Three detectors, no default labels: (1) own history: months 10-12 vs the account's months 1-9 (new missed EMIs, DPD jump of 4+ of its own SDs, 50-pt fall in share paid); (2) peer segment: robust z >= 4.5 vs loan type x income band on EMI/income, bureau utilisation, debt/income, new lines, write-offs; (3) Isolation Forest, top 1.5%. 654 accounts flagged; back-test default rate 28.4% vs 15.5% unflagged.
Trigger	Analyst submits a question.	Every build (nightly in production) over all accounts.
False positive	A confident answer on the wrong slice (e.g. a synonym mapped to the wrong column).	A harmless outlier, e.g. high EMI/income with income not captured, or a one-off missed EMI.
False negative	A valid question declined or mis-parsed.	Slow deterioration that stays inside own and peer norms; new borrowing after the last bureau pull.
Human review	Scope line, result table and query spec shown under each answer; thumbs feedback.	Analyst records Confirm / Dismiss / Need info with reason; no customer action without it; dismiss reasons retune thresholds.

Model results and approach

Model	Holdout AUC	KS	Gini	5-fold CV AUC	Out-of-time AUC (2023)
Logistic regression	0.679	0.269	0.357	0.676 ± 0.006	0.693
LightGBM (champion)	0.677	0.274	0.354	0.677 ± 0.009	0.686

35 numeric features (12-month repayment behaviour, bureau aggregates, affordability ratios, demographics, missing flags) + 9 categoricals. Collections data excluded (post-delinquency leakage). Split grouped by customer; champion by grouped-CV AUC. Both models perform alike (the signal is close to additive); LightGBM is kept for TreeSHAP account explanations, LR as benchmark. Top drivers: income, years employed, 12-month DPD, bureau DPD, age, late payments. Leakage check with bureau lines opened before origination only: AUC 0.672. Monitoring: score and feature PSI (<0.10 / 0.10-0.25 / >0.25), AUC/KS on matured cohorts, calibration by band, segment fairness, override and anomaly-dismiss rates.

Top 3 portfolio insights

1. Mortgages carry most of the risk in money terms. Mortgages are 10% of accounts but 58% of exposure and 70% of defaulted exposure. Their default rate is 24.7% against 15.2% for all other products. *So what:* Exposure-weighted, the portfolio's risk is a mortgage question. Prioritise a mortgage underwriting review and request LTV and property values, which are missing from this data, so loss given default can be estimated.

2. Income is the clearest single driver. Default rate falls steadily with income, from 25.7% below $58k to 9.5% above $101k (2.7x). The lowest band holds 20% of exposure but 30% of defaulted exposure. *So what:* Affordability should drive limits: cap EMI-to-income for the lowest band and route sub-$58k applications for larger loans to manual review.

3. Unemployed, low-income borrowers are the worst segment (cross-cut). Unemployed borrowers earning under $58k default at 38.4% (531 accounts), against 9.1% for salaried borrowers above $101k. Unemployed borrowers are 5% of exposure but 11% of defaulted exposure. Low-income mortgages default at 35.4%. *So what:* A combined policy rule (employment status x income) is sharper than either cut-off alone. Require a guarantor or income proof for unemployed applicants and review the existing low-income mortgage book first.

Key assumptions

Data is a snapshot at 2025-12-31; exposure = outstanding balance after month 12; target_default is final and its observation window is unknown; bureau is the latest snapshot (a behavioural, monitoring-style score), checked against a point-in-time variant; 7,293 loans predating customer_since, 3,875 collections events before origination and 3,149 future close dates are treated as data errors and flagged, not dropped.

What I would need from the bank's IT and data teams for production

(1) Governed feeds from core banking, the bureau and the collections system into the warehouse, with point-in-time bureau archives and a defined default definition and outcome window. (2) A scheduled pipeline (Airflow/dbt) with data-quality tests that block bad loads. (3) Model registry, validation sign-off and a scoring service. (4) An approved LLM endpoint inside the bank's cloud tenancy with no data retention, plus logging of prompts and answers. (5) SSO and role-based access for the dashboard, and a decisions table in the database instead of a CSV. (6) Model risk policy for monitoring thresholds, fairness review and retraining.